

Tessera: A Domain-Specialized Multi-Agent System for Enterprise Transformation

Architecture, Benchmarks, and the Compounding Knowledge Moat

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Abstract

Frontier large language models (LLMs) have become extraordinarily capable general reasoners, yet capability on open-ended tasks is not the same as reliable performance on a specific enterprise’s accumulated, idiosyncratic software estate. This paper presents Tessera Labs’ multi-agent system for enterprise software comprehension, migration, and orchestration, and makes three contributions. First, we introduce **TILES, our enterprise benchmark**: 42 tasks across seven representative ERP-transformation suites, each labeled *contamination-proof*, *public*, or *control* according to whether its answer can be reached without the customer’s proprietary knowledge—a design that separates genuine enterprise knowledge from pretraining recall. Second, we show that embedding a frontier model in Tessera improves task accuracy by a mean of **28.9 percentage points** over the same model used directly, and by more than **64 points on contamination-proof tasks** whose answers cannot be produced from pretraining memory, confirming the gains are attributable to knowledge rather than leakage. Third, we argue that this advantage **compounds with usage**: the base model is a commodity input, while Tessera’s persistent enterprise system model accrues firm- and time-specific knowledge that a competitor—or an improved base model—cannot shortcut. When the underlying model improves, Tessera inherits the gain and *retains* its accumulated lead.

1 Introduction

Enterprise transformation—particularly the modernization of enterprise resource planning (ERP) systems—is fundamentally a **software comprehension and systems coordination problem**. Large ERP systems such as SAP ECC evolve over decades, accumulating millions of lines of custom code, tightly coupled integrations, and business-critical workflows embedded across hundreds of modules. The difficulty of modernization does not arise from infrastructure replacement alone, but from the need to **understand undocumented custom logic** across thousands of programs, **migrate and reconcile data** across heterogeneous schemas and formats, **preserve behavioral correctness** while modifying core systems, **coordinate changes** across multiple interconnected enterprise platforms, and **test business-critical processes** across this fragmented technology stack. Gartner predicts that more than 70% of recently implemented ERP initiatives will fail to fully meet their original business-case goals [1]—not because teams lack discipline, but because the real system on disk is never the system the documentation describes.

1.1 Standard versus high-stakes knowledge work

Two capabilities determine performance on enterprise work, and they are not the same thing. The first is **intelligence**: the ability to analyze, reason, generate, and solve problems in the abstract. Frontier models now supply this in abundance. The second is **wisdom**: knowing not just what exists but why—that two systems which look independent in fact exchange data through a nightly batch job written a decade ago, and that the last change to it broke finance reconciliation. Intelligence is broadly available; wisdom about a specific enterprise is not, because it exists only inside that enterprise and is accumulated slowly, one migration, audit, and incident at a time. This maps directly onto the kind of work involved. General-purpose

LLMs excel at *standard* knowledge work: tasks that follow a common process, can be executed from publicly available information, and do not depend on proprietary context [2]—work that intelligence alone can carry. ERP transformation is the opposite. It is *high-stakes* knowledge work: it depends on deep domain expertise, on a constantly evolving and proprietary knowledge base (the customer’s own system), and it is rarely done the same way twice. No amount of reasoning substitutes for the missing context. A model asked about SAP in general can answer fluently; the same model asked what *this* customer’s fifteen years of custom ABAP actually does, and what will break if it is changed, has no grounded basis to answer reliably—not because it lacks intelligence, but because it lacks wisdom about this specific system. This distinction motivates a deliberate architectural stance, echoed across specialized AI systems in law [3], finance, and venture [4]: the frontier model is a powerful but *commodity* reasoning engine, accessible to everyone at roughly the same price per token [5]. The durable value is not the model; it is the system that embeds the model in a specific enterprise and accumulates understanding of it over time. This paper describes that system, measures the advantage it produces today, and explains why the advantage grows with use.

2 Traditional Approaches versus Tessera

Traditional transformation relies on manual analysis, static tooling, and large specialist teams. Two of its limitations are structural. The work is **fragmented**—code, data, integrations, and workflows are analyzed in isolation, so no one holds a coherent model of system behavior—and it **does not accumulate**: each project rediscovers the same undocumented logic and dependencies from scratch, leaving behind no persistent system-level intelligence. Tessera is built to remove both. It is not a general-purpose assistant but a **purpose-built multi-agent system for enterprise software comprehension, migration, and orchestration**, operating over a unified model of the customer’s code, data, and integrations. It ships pre-loaded with the knowledge of veteran ERP practitioners, and once deployed it self-adapts and evolves with every interaction—accumulating organizational understanding that persists as a reusable expert layer long after the initial transformation.

3 Tessera System Architecture

System Ingestion Layer. Connects to platforms such as SAP, Oracle, Salesforce, and Workday and converts raw artifacts—source code, schemas, configuration, and integration interfaces—into structured, machine-readable form. From these it builds a comprehensive **entity graph** [6] linking objects both *within* a system (e.g., Salesforce Accounts to Contacts via `AccountId`) and *across* systems (migration mappings, shared business processes such as Order-to-Cash, and the custom programs that touch each entity). A **process-mining agent** [7] extends the graph with the custom tables and process variants unique to the organization.

Structured Knowledge Layer. Consolidates these artifacts and the entity graph into a single **unified model** of the enterprise, with code, data, integrations, and execution relationships explicitly linked. This lets Tessera reason about system behavior, change impact, and downstream effects—and automate or reengineer workflows—in ways that flat, text-based code indexes cannot.

Domain-Expert Knowledge Base. Alongside the customer-specific entity graph, Tessera ships with a curated, structured store of enterprise IT knowledge authored by veteran practitioners and not present in general-purpose models. Each *posting* captures a discrete piece of operational knowledge—a custom-object pattern, a configuration rule, a migration pitfall—together with the conditions under which it applies and its provenance, so that agents can retrieve and apply it with traceable justification. Figure 2 shows an illustrative posting.

Multi-Agent Reasoning Layer. Decomposes each request into subtasks handled by specialized agents that collaborate over the shared model [8]. A base agent, **Tessa**, triages requests and generates structured

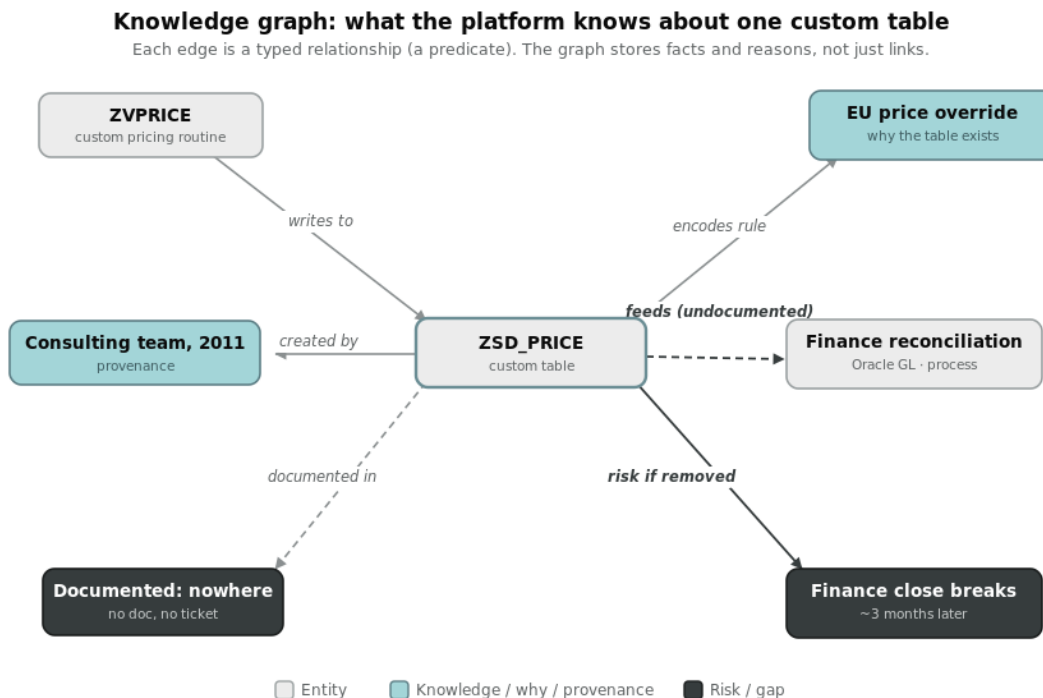


Figure 1: An illustrative slice of the entity graph: what Tessler links about a single custom table (**ZSD_PRICE**). Nodes capture not only the entities but also the rules that explain why they exist and the risks they carry; each edge is a typed relationship. This structured, reasoned context is precisely what a raw model has no access to.

execution plans—which downstream agents to invoke, with what objective and parameters—while sub-agents handle data processing, application operations, integrations, and code analysis. Agents combine a *short-term memory* of the current session with an *organizational memory* drawn from the persistent system model, and every action further refines the entity graph, so institutional intelligence accumulates over time.

Execution and Validation Layer. Turns plans into real-world actions through **deterministic pipelines**: tools validate inputs and verify readiness before any change, and all actions are **auditably tracked**. This combination of traceability, explainability, and human-review gating is what makes the system’s output acceptable to enterprise risk officers.

4 TILES: Transformation & Integration of Legacy Enterprise Systems Benchmark

4.1 Motivation and design

Public LLM leaderboards measure general reasoning; they do not measure whether a system can safely comprehend and transform a specific enterprise’s accumulated software estate. Following the spirit of domain-specific evaluations such as Harvey’s BigLaw Bench [3] and Aida’s internal benchmarks in venture [4], we constructed **TILES (Transformation & Integration of Legacy Enterprise Systems Benchmark)** targeting the tasks that actually consume transformation budgets. The benchmark comprises **seven task suites of six tasks each (42 in total)**, drawn from real engagement work: **SAP ABAP Coding**, **SAP Configuration Analysis**, **Finance Schema Modeling**, **Manufacturing Schema Modeling**,

KB Posting

Entry: MKPF / MSEG—material-document tables under S/4HANA

Domain: SAP MM-IM (Inventory Management), S/4HANA conversion

Applies when: custom ABAP issues a direct INSERT / UPDATE / DELETE on MKPF or MSEG, or reads aggregated stock from MARD / MARC / MCHB

Knowledge: In S/4HANA the material-document data model moved to the denormalized MATDOC table. MKPF and MSEG remain as DDIC objects but are served by CDS compatibility views (NSDM_DDL_MKPF, NSDM_DDL_MSEG): reads are transparently redirected, but writes are not—a direct INSERT into MSEG silently has no effect, so goods movements vanish without an error. Aggregated stock in the hybrid tables is computed on the fly from MATDOC.

Migration note: replace direct writes with the released BAPI_GOODSMVT_CREATE; read stock through the NSDM access classes (e.g., CL_NSDM_SELECT_MARD) rather than selecting on the compatibility view.

Provenance: SAP Note 2206980 (MM-IM data-model change in S/4HANA); related material-number handling in SAP Note 2267140.

Status: verified against SAP Notes 2206980 and 2267140

Figure 2: A domain-expert knowledge-base posting, grounded in SAP Notes 2206980 and 2267140.

Cross-System Reconciliation, Tabular Data Prediction, and Test Requirement & Test-Data Generation.

Task classes. Alongside its suite, each task is labeled by class, based on whether its answer depends on the customer’s own systems or on general knowledge. This is what makes the benchmark a real test of what Tessera adds, not of what the model already knows. **(i) Contamination-proof** tasks can only be answered from inside the customer’s environment. The facts are not in any public source, so a model working on its own has nothing to go on. One task, for instance, asks how a tenant-specific field maps between two of the customer’s systems, while another asks what a custom ABAP program will break when its underlying tables change. **(ii) Public** tasks rely on standard SAP or ERP knowledge that is publicly available. They require enterprise domain expertise, but it is knowledge that frontier models already have access to. Examples include how SAP’s standard sales-order tables relate and what a standard delivery-class setting controls. **(iii) Control** tasks do not need Tessera at all. Everything required is already in the prompt—cleaning malformed rows in a table we provide, say, or transforming data whose rules are spelled out in the question. They are used as a check: if the Tessera-embedded model improves on these, something is wrong. We expect no real lift here, and that is what we find.

Grading. Each task is graded against an expert-authored rubric that decomposes a correct response into discrete, verifiable claims; “task accuracy (%)” is the rubric-weighted fraction satisfied. Open-ended tasks are scored by a **three-judge LLM ensemble** using a **2-of-3 majority vote**. Structured tasks use a deterministic check, and hybrid tasks average the two. The judge panel is disjoint from the candidate models (no model grades itself). Inter-judge agreement averages ~ 0.90 , and results are robust to the aggregation rule (mean-of-judges differs by ≤ 1 point; §5.4). Each cell is run five times to average out sampling noise.

4.2 Baselines and configurations

Tessera is model-agnostic by design: its reasoning engine is abstracted behind the agent harness, which can route across frontier models. To isolate the contribution of Tessera-embedding from that of the underlying model, we evaluate five frontier models—**Claude Opus 4.8**, **Claude Sonnet 5**, **GPT-5.4**, **Gemini 3.5 Flash**, and **Claude Haiku 4.5**—in two configurations each ($42 \text{ tasks} \times 5 \text{ models} \times 5 \text{ runs} = 1,050 \text{ runs per configuration}$):

- **Raw frontier model (baseline):** the model invoked directly on the task, with access to the same source artifacts but not embedded in Tessera.
- **Tessera-embedded:** the identical model with Tessera’s domain knowledge supplied in context, holding

the reasoning engine fixed.

The difference between the two is what we call the **Tessera-embedding lift**: the accuracy attributable to Tessera’s domain knowledge and specialization.

5 Results

Across all seven task suites and all five models, embedding the model in Tessera improves task accuracy by a mean of **+28.89 points** over baseline ($p < 10^{-5}$; Table 1, Figure 3). The effect is not confined to any one model: Tessera improves *every* frontier engine we tested, which is the central point—it is Tessera, not the model, that closes the gap to an expert practitioner. On absolute Tessera-embedded accuracy, Claude Opus 4.8 leads all engines (90.29), with Claude Sonnet 5 close behind (89.92).

Table 1: Task accuracy (%): raw frontier model $\rightarrow$ Tessera-embedded, averaged over per-task means (majority-vote grading; unparseable/refusal = 0). Lift is the matched-pair difference; columns are rounded independently, so Lift may differ from the column difference by ≤ 0.01 . Final row is the mean over models.

Backbone	Baseline	Tessera-embedded	Lift
Claude Opus 4.8	65.92	90.29	+24.37
Claude Sonnet 5	63.02	89.92	+26.91
GPT-5.4	59.59	89.33	+29.75
Gemini 3.5 Flash	59.13	84.40	+25.27
Claude Haiku 4.5	43.72	81.87	+38.15
Mean	58.27	87.16	+28.89

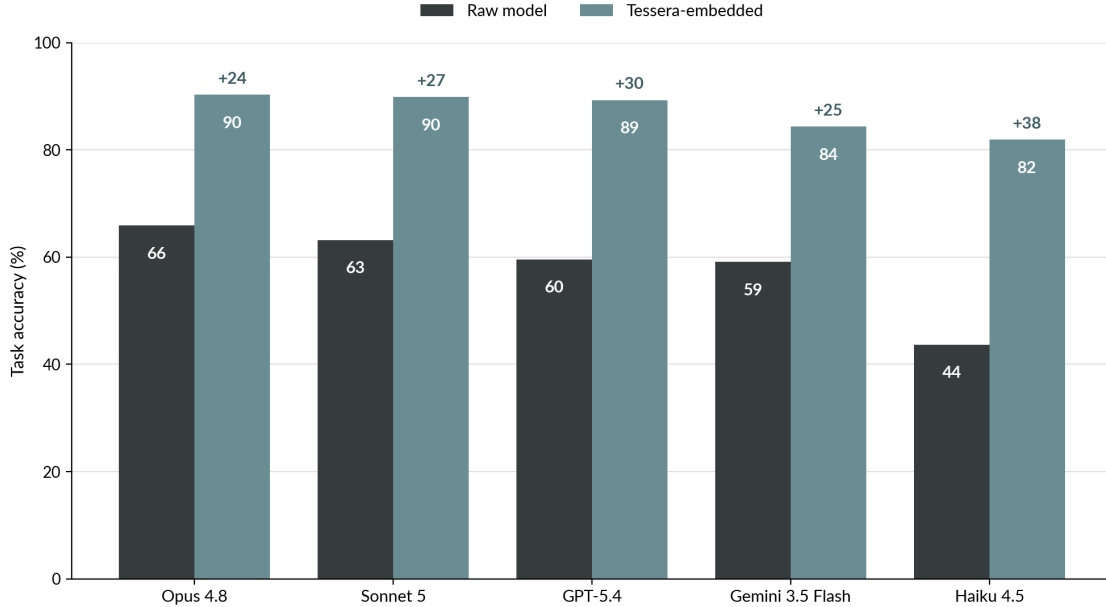


Figure 3: Task accuracy for each model, raw versus Tessera-embedded. Labels show the lift in points.

5.1 Validity: embedding helps most where it cannot be faked

The most important check on a benchmark like this is whether the measured lift reflects real, proprietary knowledge or mere leakage. The task-class breakdown answers this directly (Table 2, Figure 4). On **contamination-proof** tasks—whose answers exist only in the customer’s own systems—embedding adds +64.32 **points** ($p < 10^{-5}$), the largest effect in the study. On **public** tasks the lift is smaller but still significant (+18.06; $p = 0.009$). On **control** tasks the lift is *not significantly different from zero* (+0.53; n.s.)—exactly the desired result: Tessera is not manufacturing lift on tasks that should not need it. Together these establish that the gains are attributable to knowledge, not contamination or a grading artifact.

Table 2: Tessera-embedding lift by task class, in points over baseline, with per-task paired significance (p).

Task class	Lift	Significance
Contamination-proof	+64.32	$p < 10^{-5}$
Public	+18.06	$p = 0.009$
Control	+0.53	n.s.

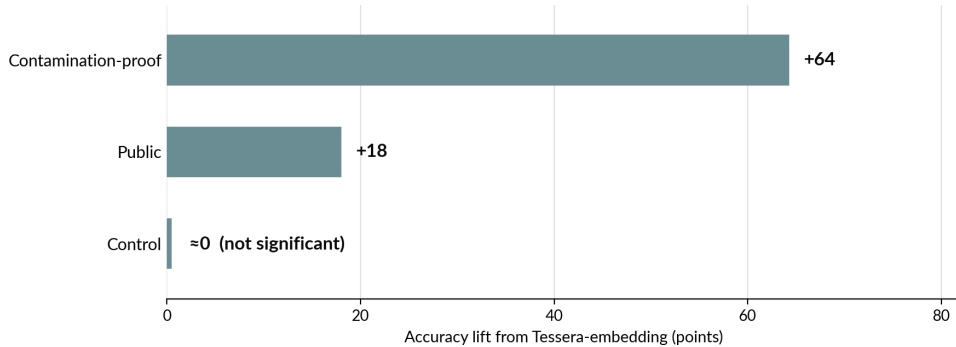


Figure 4: Tessera-embedding lift by task class.

5.2 Results by task suite

The lift is not uniform across task types, and the pattern is informative (Table 3, Table 4). On **code** tasks (ABAP coding, configuration analysis) raw models already reach moderate accuracy, because much of the task is self-contained in the artifact in front of them; Tessera’s contribution is dependency awareness and adherence to the customer’s standards. On **data and process** tasks correctness depends on relationships not visible in any single artifact—this is where Tessera’s embedded enterprise knowledge contributes most, with the very large lifts on tabular prediction (+63.33) and cross-system reconciliation (+31.93). (Tabular scores fall on round fractions because those tasks are exact-match, scored 0 or 100 per item.) The lift is not uniform even within a suite: on manufacturing schema modeling, where raw accuracy is already near ceiling, embedding lowers the strongest model (Opus, 98.75 → 90.83) while raising the weaker ones—a sign that retrieved context can distract when a model already knows the answer. **Testing** draws on both and shows one of the largest combined lifts (+45.53). The throughline is that model capability and Tessera’s knowledge are complementary inputs, not substitutes: Tessera supplies the enterprise-specific relationships, and the model supplies the reasoning that turns them into correct work.

Table 3: Task accuracy (%) by suite: raw frontier model $\rightarrow$ Tessera-embedded, mean over the five models. Lift is the matched-pair difference; columns are rounded independently.

Task suite	Baseline	Tessera-embedded	Lift
Tabular Data Prediction	36.67	100.00	+63.33
Test Req. & Test-Data Gen.	41.04	86.57	+45.53
SAP Configuration Analysis	48.88	82.71	+33.84
Cross-System Reconciliation	66.68	98.61	+31.93
Finance Schema Modeling	68.27	80.56	+12.29
SAP ABAP Coding	61.80	69.61	+7.80
Manufacturing Schema Modeling	84.59	92.08	+7.49
Mean	58.27	87.16	+28.89

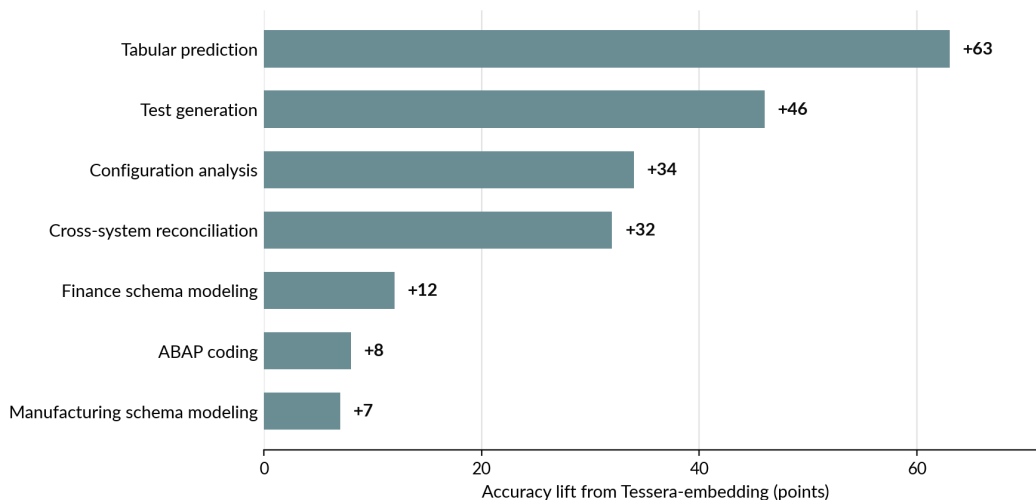


Figure 5: Tessera-embedding lift by task suite (Tessera-embedded vs. baseline, mean over models).

5.3 Why the model still matters

The headline result—that embedding lifts *every* model—should not be read to mean the base model is interchangeable. Embedding produces a large, significant lift for every engine (+24 to +38 points), and two patterns are informative. First, the largest lift accrues to the weakest starting point: Claude Haiku 4.5, the smallest and least expensive model, gains +38.15 points—a floor effect—showing embedding can carry an inexpensive model most of the way, though it remains the lowest on absolute accuracy. Second, model choice sets the *ceiling*: Claude Opus 4.8 reaches the highest absolute Tessera-embedded accuracy (90.29), with Sonnet 5 close behind (89.92), because a stronger reasoner exploits the same retrieved knowledge more fully. The “commodity” framing is therefore about *access*, not interchangeability: reasoning is broadly available at similar cost per token, but model choice still determines the achievable ceiling, and Tessera is deliberately designed to run on the strongest available engine.

5.4 Grading reliability and robustness

Inter-judge unanimity—the fraction of rubric claims on which all responding judges agreed—averages 0.894 (baseline) and 0.902 (Tessera-embedded), supporting the reliability of the ensemble. The headline results

Table 4: Detailed task accuracy (%): baseline $\rightarrow$ Tessera-embedded, by suite and model. Final row is the mean lift across suites.

Task suite	Opus 4.8	Sonnet 5	GPT-5.4	Gemini 3.5	Haiku 4.5
<i>Code</i>					
SAP ABAP Coding	76.79 $\rightarrow$ 76.91	70.01 $\rightarrow$ 77.87	68.62 $\rightarrow$ 71.57	48.90 $\rightarrow$ 65.48	44.71 $\rightarrow$ 56.20
SAP Configuration Analysis	46.60 $\rightarrow$ 85.75	56.75 $\rightarrow$ 88.89	45.06 $\rightarrow$ 87.42	55.02 $\rightarrow$ 85.19	40.95 $\rightarrow$ 66.31
<i>Data & process</i>					
Finance Schema Modeling	79.41 $\rightarrow$ 84.64	70.08 $\rightarrow$ 82.73	72.82 $\rightarrow$ 82.73	65.00 $\rightarrow$ 76.07	54.03 $\rightarrow$ 76.63
Manufacturing Schema Modeling	98.75 $\rightarrow$ 90.83	88.93 $\rightarrow$ 89.58	87.08 $\rightarrow$ 92.50	87.38 $\rightarrow$ 93.75	60.83 $\rightarrow$ 93.75
Cross-System Reconciliation	74.88 $\rightarrow$ 100.00	72.25 $\rightarrow$ 97.22	66.11 $\rightarrow$ 100.00	66.66 $\rightarrow$ 96.94	53.47 $\rightarrow$ 98.89
Tabular Data Prediction	33.33 $\rightarrow$ 100.00	33.33 $\rightarrow$ 100.00	33.33 $\rightarrow$ 100.00	50.00 $\rightarrow$ 100.00	33.33 $\rightarrow$ 100.00
<i>Testing</i>					
Test Req. & Test-Data Gen.	51.69 $\rightarrow$ 93.89	49.76 $\rightarrow$ 93.17	44.09 $\rightarrow$ 91.10	40.95 $\rightarrow$ 73.35	18.71 $\rightarrow$ 81.35
Mean lift (pts)	+24.37	+26.91	+29.75	+25.27	+38.15

are robust to the aggregation rule: under a mean-of-judges aggregation the lift is +28.6 points, within ~ 1 point of the majority-vote figure, with every per-suite and per-class entry within ~ 1 point and no systematic bias.

6 Limitations

This is a preliminary benchmark and we flag its bounds explicitly. (i) *Scale and scope*: 42 tasks across seven suites drawn from a single tenant’s engagement data; broader coverage and additional tenants are future work. (ii) *Grading*: scores rely on an LLM-judge ensemble; although grading is rubric-based and inter-judge agreement is high (~ 0.90), judge bias cannot be fully excluded.

7 The Compounding Knowledge Moat

The benchmark measures Tessera’s advantage *today*. The more important question for an enterprise committing to a multi-year transformation is whether that advantage survives the relentless improvement of the underlying models. It does—and it widens—because of where the value actually lives.

7.1 The model is the commodity; the system is the moat

Infrastructure and model access create little durable advantage; they can be replicated quickly and are commoditized by the arms race between labs [5]. Specialization is the axis on which moats are built, and it compounds in one direction: inward, toward the specific customer. It is useful to think in deepening layers of enterprise knowledge:

- **Domain depth**—understanding the field (ERP, ABAP, S/4 migration patterns). This is the layer Tessera ships with on day one, and it is what the benchmark measures. Necessary, but replicable by anyone with access to the same domain expertise.
- **Firm depth**—understanding *this* enterprise: its custom tables, its undocumented logic, its integration topology, its process variants. Tessera acquires this as it ingests and mines the customer environment.

- **Personal and temporal depth**—understanding that accumulates across sessions: which recommendations were accepted or overridden, which risks materialized, how the customer’s engineers actually work. This is observed and inferred over time, never explicitly taught.

Tessera’s **persistent enterprise system model** is precisely the architecture that captures these deeper layers. Unlike a stateless, request-response assistant that forgets between sessions, Tessera is a stateful, long-running system: every interaction is a deposit into an asset that becomes more valuable and harder to abandon. The switching cost of a stateless tool is near zero; the switching cost of a system that has spent two years learning a specific SAP estate is enormous—not because of contractual lock-in, but because of what would be lost.

8 Conclusion

Enterprise modernization and automation are fundamentally challenges of system comprehension and coordinated execution. Tessera addresses them through a purpose-built architecture that combines structured system modeling, multi-agent reasoning, and deterministic, audited execution. Our benchmark shows that embedding a frontier model in Tessera lifts task accuracy by a mean of **28.9 points** across every model tested (and by more than **64 points on contamination-proof tasks**), a statistically significant effect that is consistent across engines and absent on control tasks—enough to materially change the economics of a transformation. Because the value lives in a persistent enterprise system model rather than in the commodity reasoning engine, this advantage compounds with usage and is reinforced, not eroded, by progress in the underlying models. The common fear that a frontier lab will release a model that obsoletes a specialized system rests on a false premise—that the system competes with the model. Tessera does not: it is built to *absorb* model improvements, so when a stronger base model ships, every deployment gets better the same day, on top of the firm- and time-specific knowledge already accumulated. An enterprise adopting Tessera is therefore not betting on which lab wins; it is investing in a system whose advantage is denominated in *accumulated, audited usage over time*—the one quantity a better model cannot retroactively supply.

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